

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

30th October 2015

Subject CT2 – Finance and Financial Reporting

Time allowed: Three Hours (10.30 – 13.30 Hrs.)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. Please read the instructions on the front page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
- 2. Mark allocations are shown in brackets.*
- 3. Attempt all questions, beginning your answer to each question on a separate sheet. However, answers to objective type questions could be written on the same sheet.*
- 4. Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately.

- Q. 1)** Company ABC has given a loan of INR 20 Million to Company XYZ. The first installment of INR 6 Million repayment is due on 31st March 2015 which includes INR 5 Million as capital installment and INR 1 Million as interest payment.

Which of the following situations would imply credit risk for Company ABC?

- I.** No payment is made due to the bankruptcy of Company XYZ.
- II.** The payment is rescheduled – INR 5 Million will be paid on 31st March and INR 1 Million will be paid on 15th April
- III.** Company XYZ is not bankrupt, but the management has decided not to pay full interest and hence the total payment made will be INR 5.5 Million
- IV.** The payment is not made due to fraud by an employee in Company XYZ

- A)** Only I
- B)** I, II & III
- C)** All – I, II, III & IV
- D)** None of the above

[2]

- Q. 2)** The market price of each share of Bahubali Ltd is INR 200. The face value of each share is INR 20. The capital gearing ratio is 0.5. The Price Earnings Ratio is 20. The earnings from ordinary activities for the immediately preceding year were INR 2,000 million. The value of its borrowings is

- A)** Impossible to find with the available information
- B)** INR 20,000 million
- C)** INR 4,000 million
- D)** INR 2,500 million

[2]

- Q. 3)** Under which of the following circumstances will a company opt for recourse factoring?

- A)** When the credit worthiness of the customer is not known
- B)** When the company is in need of financing expertise for recovery from the customers
- C)** When the company wants to earn good rate of interest income
- D)** When the company is in need of advance payment to manage its working capital

[2]

- Q. 4)** Which of the following derivative positions (assuming options are exchange traded) require payment of margins?

- I.** Long position on a futures contract
- II.** Short position on a futures contract
- III.** Short position on a call option
- IV.** Long position on a put option

- A)** II only
- B)** All of the above
- C)** I, II, and III.
- D)** I, II, and IV.

[2]

- Q. 5)** Which of the following leads to an immediate increase in the number of equity shares and also an increase in the amount of equity
- I.** Placing
 - II.** Scrip issue
 - III.** Right Issue
 - IV.** Introduction
- A)** I and III.
B) All of the above
C) I, II, and III.
D) I, III, and IV. [2]
- Q. 6)** Which of the following errors in recording transactions has an impact on profitability?
- I.** Purchase of a new machinery in the last month of the financial year was not recorded at all
 - II.** Cash paid to one of the suppliers recorded as being paid to buy a depreciable asset
 - III.** Loan taken from bank at the start of the year not being recorded at all
 - IV.** Amount collected from one of the customers to whom goods were sold on credit being recorded as additional sales made
- A)** IV. only
B) All of the above
C) I, II, and IV.
D) II and IV. [2]
- Q. 7)** Which of the following are correct for sale and leaseback?
- I.** It allows the owner of the asset to release the capital tied in the asset
 - II.** It mandates the original owner of the asset to buy back the asset at the end of the period
 - III.** The opportunity to gain from future appreciation of the asset is lost
 - IV.** Sale and Leaseback can be executed on depreciable as well as non-depreciable assets
- A)** I only
B) III only
C) I, II, and IV.
D) I, III, and IV. [2]
- Q. 8)** Which of the following would be examples of specific risk for a large domestic house building company?
- A)** Inflation
 - B)** Interest rates
 - C)** Regional variations in house prices
 - D)** Recession [2]

Q. 9) Reliant Ltd has decided to make a Bonus issue. Its share capital stands at Rs. 100,000 (Face Value = Rs. 10) and its reserves are Rs. 50,000. Its share price is Rs. 21. It makes a 1-for-2 Bonus issue. What is the theoretical effect on the share price and what will be the share capital and reserves after the Bonus issue?

- A) Rs. 21; Rs. 100,000; Rs. 50,000
- B) Rs. 42; Rs. 200,000; Rs. 50,000
- C) Rs. 10.5; Rs. 125,000; Nil
- D) Rs. 14; Rs.150,000; Nil

[2]

Q. 10) Which of the following would NOT be included in a firm's equity?

- A) Retained earnings
- B) Dividends
- C) Share capital
- D) The upward revaluation of a non-current asset

[2]

Q. 11) Rohit Ltd has provided you with the following balances as on 31st March 2015

	Amounts in INR millions
Printing and stationery	12
Wages	3500
Salary to administrative staff	400
Sales	20000
Purchases	11500
Trade receivables	3000
Trade payables	2250
Insurance premium	15
Factory land	1005
Factory building	2400
Motor vehicles	40
Cash and bank balances	554
Equity share capital	1000
12% Preference share capital	500
8% Secured debentures	4000
Investments	1150
Machinery and equipment	5650
Power and fuel expenses	1450
Distribution costs	750
Advertisement expenses	250
Inventory	2450
Reserves and surplus	6950
Bills receivable	450
Prepaid wages	124

Notes:

- 1) Value of inventory of raw material and finished goods was INR 1,600 million as at 31st March 2015.

- 2) Depreciation needs to be charged on the depreciable non-current assets at the following rates:

	Rate of Depreciation p.a.
Factory building	10%
Motor vehicles (used by Sales Managers)	20%
Machinery and equipment	20%

- 3) Prepaid wages of INR 124 million included in the list of balances above are as at 31st March 2014.
- 4) The 12% Preference share capital and 8% Secured debentures were issued on 1st April 2012 and are redeemable 10 years from the date of issue.
- 5) On 1st March 2015 bills receivable of INR 40 million were endorsed to a supplier of raw materials in lieu of goods supplied by him. The endorsed bill would mature on 31st May 2015. This transaction was appropriately recorded in the books. However on 31st March 2015, the supplier requested that he be paid in cash immediately and that he would return the bills of exchange endorsed to him on 1st March 2015. After negotiation, it was agreed that he will receive 97.5% of the amount in cash immediately in full and final settlement of the dues of INR 40 million. The necessary transaction was executed just before closure of books on 31st March 2015 but was not yet recorded in the books of accounts.
- 6) Prepaid wages as at 31st March 2015 stood at INR 100 million.
- 7) Outstanding salary to administrative staff as at 31st March 2015 stood at INR 40 million.

You have been requested to prepare:

- i) Statement of comprehensive income for the year ended on 31st March 2015. (10)
- ii) Statement of financial position as on 31st March 2015 (13)
- [23]**

- Q. 12)** In the country of Centuria, gains made on financial assets that are sold within 1 year of purchase are taxed at a flat rate of 20% while gains on financial assets which are sold after being held for more than 1 year are taxed at 10%. The tax laws of Centuria prohibit any losses on sale of financial assets to be carried to the subsequent years. However the losses incurred on sale of assets could be offset against gains made in the same year. Mr. Spooky, a citizen of Centuria holds 13322 shares of Global Ltd which he bought on 01st February 2014 at a price of INR 250 per share. The market price of each share of Global Ltd. today i.e. on 15th January 2015 is INR 319.75.

Global Ltd is expected to publish its financial statements for the year 2014 by 31st January 2015. There are several contradicting rumours in the market about the expected financial results of Global Ltd for the preceding year. Adverse results are anticipated. Based on these rumours, the price as at 1st February 2015 is expected to be either of the following

INR 390 with a 40% chance
INR X with a 60% chance

Ignore the transaction costs and time value of money

- i) What is the value of X (price of shares of Global Ltd on 1st February 2015 that has a likelihood of 60%) so that either of the alternative timings of sale are financially equal? (3)
 - ii) If the value of X is expected to be higher than the one calculated in i) above, what should Mr Spooky's decision be? (1)
 - iii) What should be the price of each share today if X is equal to 240 so that either of the alternative timings of sale are financially equal? (2)
 - iv) List 4 other factors that should be considered while making a decision. (2)
 - v) Suggest 1 way in which the risk of adverse movement in share price can be eliminated and the key benefit, drawback and challenge in doing so (3)
- [11]**

Q. 13) Pujara Pvt. Ltd has accumulated large amounts of unwanted cash with no investment opportunities available as it operates in a declining but currently highly profitable industry. It is not listed on the stock exchange. It has only two key shareholders who own nearly 90% of the equity share capital but the Board comprises mostly independent professionals as these shareholders don't want to get involved in the management of the company. Its Board is of the opinion that the gearing of the company needs to be increased to make the taxation benefits of debt capital more prominent. The Board has also expressed concerns over the dim future of the industry and future profitability. The shareholders have indicated that they have alternative investment opportunities available which have a much brighter long term prospect.

- i) Identify what the Board of Pujara Pvt Ltd. should do, and indicate how this proposed course of action would meet the desired objectives (2)
 - ii) What are the different ways in which this could be implemented? (3)
 - iii) What is the key factor that the minority shareholders should consider while participating in the solution provided in i) above? (1)
- [6]**

Q. 14) XYZ insurance company is evaluating whether or not to re-insure its liability insurance portfolio. It has stopped writing business and is currently running off its reserves (i.e. paying off the claims that will arise from the business till exhaustion without any new business inflow).

The actuarial estimate of the claims to be paid in the first year is Rs. 40 Million, for the next few years it is expected that the claims payout will be 62.5% of the previous year's payment. Finally in the 6th year the balance of claims will be paid off. The total of claims paid over these 6 years is Rs. 100 Million. The claims payments due within a year are expected to be paid evenly throughout the year.

The company holds undiscounted reserves (i.e. Rs. 100 Million) in its balance sheet to pay off these claims.

The company earns 9% investment returns on the balance available on the reserves after payment of claims evenly throughout the year. It can be assumed that the interest earned/credited at the end of the year is on the opening balance less half of claims paid during the year.

A reinsurer, PQR Ltd, has given a quote of Rs. 90 Million to take over this portfolio. The terms of this transaction are as follows:-

- 1) A deposit premium of 15% of the premium will be paid up front.
 - 2) Out of the balance of the premium, a fund will be created out of which the claims will be paid off and interest will be credited at the end of the year based on the assumption mentioned above. In case the fund goes negative, then there will be negative interest credit to consider advance payments made by the insurer.
 - 3) The balance of the fund at the end of 6th year will be paid or recovered from the reinsurer based on the positive or negative balance of the fund.
 - 4) The interest rate for the above transaction will be 9%.
- i) Calculate the net present value of the XYZ's cashflows if it chooses not to reinsure it's liability insurance portfolio (5)
- ii) Calculate the net present value of the XYZ's cashflows and PQR's cashflows if XYZ chooses to reinsure it's liability insurance portfolio. Also state whether XYZ should reinsure it's liability insurance portfolio by comparing the NPV for XYZ as calculated in sub-part i) above with the NPV calculated in this sub-part (7)
- iii) If the company XYZ wants to achieve the same level of profits with or without reinsurance, then what should be the negotiated reinsurance premium? (2)
- iv) The CFO is not comfortable with the discounting method used and has asked you to rework the analysis using only realistic cash flows in the following format –

RI premium paid	(Rs. Xxx Million)
Add Total Claims Paid recovered	Rs. Xxx Million
Less Total Interest credited	(Rs. Xxx Million)

Profit/Loss to insurer	Rs. Zzz Million

According to him there is a positive gain to the insurer from the reinsurance transaction compared to being without re-insurance.

You are required to complete the CFO's analysis in the above format. And comment on the suitability of the format. (2)

- v) Make necessary corrections in sub-part iv) above by adding some more element to the format to reconcile with the answer you got in the sub-part ii) above. (6)
- [22]

- Q. 15)** Company A and B are identical companies except for their capital structure. The capital structure of each company is as follows:-

<u>Company A</u>	
Equity	400 Million
8 % Bank Loan	40 Million
9 % Debentures	160 Million
11.2% Unsecured Loan	200 Million
<u>Company B</u>	
Equity	300 Million
7% Bank Loan	150 Million

Prevailing tax rates in the country are at 40%.
 Risk free interest rates can be considered at 6%.
 Market expected returns are at 12%.

Beta of Company B is 1.5

- i) State the Modigliani & Miller proposition on the Weighted Average Cost of Capital (WACC) of two identical companies (1)
 - ii) Calculate the Beta for Company A under the Modigliani & Miller assumption (5)
 - iii) Company A wants to bring its beta equal to company B's beta of 1.5. How much additional equity should it raise? (4)
- [10]**

- Q. 16)** Define the following terms along with appropriate examples.

- i) Liquidity Preference (4)
 - ii) Financial Gearing (4)
- [8]**
